COURSE DESCRIPTION

Course Code (Title)	MTH 111 (Analytical Geometry & Calculus I)						
Prerequisite	None						
Credits, ECTS, and Contact hours	Credits [3] – ECTS [5] – Contact hours [4]						
Required, elective or selective elective?	Not Required						
Course coordinator	Dr. Mohamed Wafa						
Weekly teaching-hours	Lecture [2] – Tutorial [2] – Labs [0]						
Textbook, title, author, and year	J. R. Hass , C. E. Heil, and M. D. Weir , Thomas Calculus: Early Transcendentals,14th Edition , Pearson.2017						
Catalog Description:	Concepts and theorems of differential calculus, elementary functions with emphasis on trigonometric, hyperbolic functions and their inverses, techniques of differentiation, limits, L'Hôpital's rule and indeterminate forms, the basic concepts of plane analytic geometry particularly conic sections, introduction to three-dimensional coordinate geometry and different coordinate systems, and introduction to functions of several variables and their partial derivatives, sequences, infinite series, convergence and divergence test for series, and power series expansion						
Outcomes of instruction	• Recognize the graphs of transcendental functions (trigonometric, inverse trigonometric, exponential, logarithmic, hyperbolic, and inverse hyperbolic function) and find their domain and range. • Apply different differentiation techniques to find the derivatives of transcendental functions. • Compute limits of indeterminate forms by using L'Hopital's Rule • Identify conic sections and evaluate their respective parameters. • Identify three coordinate systems (cartesian, cylindrical and spherical) for describing the location of a point in three-dimensional space and perform coordinate transformations between them. • Understand the basic concept of multivariable functions and their partial derivatives. • Select the suitable test for the convergence and divergence of infinite sequences and series.						
Rating of the Addressed Student Outcomes [H, M, L, 0]	SO1	SO2	SO3	SO4	SO5	SO6	SO7
	H						
Topics covered	basic concepts in pre-Calculus Inverse trigonometric functions Hyperbolic and inverse hyperbolic functions Indeterminate forms and L'Hopital's Rule Taylor and Maclaurin series The equation and graph of circle, parabola The equation and graph of ellipse and hyperbola Partial derivatives and applications Infinite sequences and series Coordinate systems in three-dimensional space						

* [H, M, L, 0] == High, Medium, Low, and No